Three random decks: the one-page version

What the activity is

Three little decks of number cards beat each other in a circle, like rock, paper, scissors: A beats B, B beats C, and C beats A. Children play short games with the decks, then list every pair of cards that could come up to find out which deck is really ahead.

Make the decks (once, about 30 minutes; keep them)

- Each table gets three paper lunch bags labelled **A**, **B** and **C**, each holding three cards made from half index cards, number written large:

  Bag A 2 4 9
  Bag B 1 6 8
  Bag C 3 5 7
- K–1 table: also draw each number's dots in a ten-frame (two rows of five), as on page 1 of the K–1 packet.
- Spares: one more copy of each of the nine cards per table, in pencil (later problems copy or rewrite cards). Grades 2–3 table: also cards 0, 10, 11 and 12, and the letters A, B, C on sticky notes rather than on the bags (Problem 3 hides them). The room needs 58 cards.
- K–1 table: 18 counters, no wider than 1/2 inch. Everyone: pencils and blank paper.

Launch (one minute, whole group)

Hold up bags A and B. Without looking, draw one card from each; the bigger number wins the round for its deck. Put both cards back and shake. Ask: *"How could we find every pair of cards that could come up, even ones we never draw today?"* Don't say which deck wins.

What Problem 1 is for

- **K–1:** partners play three games, A against B, B against C and C against A, six rounds each, putting a counter in the winning deck's row. The games raise the question "is one deck best?" They can't settle it: the stronger deck wins a six-round game only about 45% of the time. Accept any result. Problems 2–4 settle it by circling the winner of all nine card pairs.
- **Grades 2–3 and 4–5:** the "pick first or pick second" game, ten rounds. Each row of the table is one game: write both deck letters and their rounds won, like the shaded example row ("A, 4 wins" and "C, 6 wins"). The idea to reach: whoever picks second can always take the deck that beats the first pick. Ten rounds often won't show it; Problem 2 proves it by counting pairs.

The surprise

Each deck beats the next one on 5 of its 9 card pairs, so no deck is best, even though every deck adds up to 15.

Key answers

- **K–1.** P2–4: A beats B, B beats C, C beats A, each 5 pairs to 4. P5: no; the stronger deck can still lose six rounds. P6: only 9. P7: five swaps work, the whole row of B's 1 and the whole column of A's 9. P8: impossible, since nine pairs can't split into two equal piles. P9: impossible, since the deck holding the 1 loses at least two of its four pairs to every other deck.
- **Grades 2–3.** P2 as above. P3: no number of rounds makes the guess certain. P4: A 1, 5, 9 / B 3, 4, 8 / C 2, 6, 7, and A 2, 3, 9 / B 1, 6, 8 / C 4, 5, 7. P5: 2 or 4. P6: doubling every card changes nothing; with the four-card decks C beats both others, so the circle breaks. P7: e.g. A 3, 5, 7 / B 2, 4, 12 / C 1, 9, 11. P8: impossible, as K–1 P9.
- **Grades 4–5.** P2: 5/9 each way round; equal totals of 15 don't decide it. P3: exactly the three arrangements above. P4: e.g. A 1, 4, 4 / B 3, 3, 3 / C 2, 2, 5. P6: a two-card deck wins when its small card beats their small card and its big card beats their big card. P7: three cards per deck is the fewest. P8: impossible.